

## 6.2 The subspaces we need

This section gathers the vector-space facts used in the rest of the chapter. Readers who have worked through [Chapter 1](#) can skim the definitions. The results on orthogonal complements, especially [Lemma 6.2.1](#) and [Theorem 6.2.3](#), are used constantly afterwards and are proved in full.

### 6.2.1 Subspaces, spans and dimension

A nonempty set  $\mathcal{S} \subseteq \mathbb{R}^n$  is a **subspace** if it is closed under addition and scalar multiplication:  $\mathbf{u}, \mathbf{v} \in \mathcal{S}$  and  $a, b \in \mathbb{R}$  imply  $a\mathbf{u} + b\mathbf{v} \in \mathcal{S}$ . Every subspace contains  $\mathbf{0}$ . The **span** of vectors  $\mathbf{v}_1, \dots, \mathbf{v}_k$  is the set of all their linear combinations. It is the smallest subspace containing them. A set of vectors is *linearly independent* if the only linear combination equal to  $\mathbf{0}$  is the one with all coefficients zero. A **basis** for  $\mathcal{S}$  is a linearly independent set that spans  $\mathcal{S}$ . All bases of  $\mathcal{S}$  have the same number of elements, the **dimension**  $\dim \mathcal{S}$ .

For an  $n \times p$  matrix  $\mathbf{A}$  there are two subspaces we use constantly:

$$\begin{aligned}\mathbb{C}(\mathbf{A}) &= \{\mathbf{A}\mathbf{b} : \mathbf{b} \in \mathbb{R}^p\} \subseteq \mathbb{R}^n && \text{(column space),} \\ \mathcal{N}(\mathbf{A}) &= \{\mathbf{b} \in \mathbb{R}^p : \mathbf{A}\mathbf{b} = \mathbf{0}\} \subseteq \mathbb{R}^p && \text{(null space).}\end{aligned}$$

The **rank** of  $\mathbf{A}$  is  $\text{rank}(\mathbf{A}) = \dim \mathbb{C}(\mathbf{A})$ . Two standard facts from [Chapter 1](#): row rank equals column rank, so  $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}\mathbf{T})$ ; and the rank–nullity theorem

$$\text{rank}(\mathbf{A}) + \dim \mathcal{N}(\mathbf{A}) = p. \quad (6.2.1)$$

In regression,  $\mathbb{C}(\mathbf{X})$  is the **model space**.  $\mathcal{N}(\mathbf{X})$  is the set of directions in which  $\boldsymbol{\beta}$  can move without changing  $\mathbf{X}\boldsymbol{\beta}$ . It is  $\{\mathbf{0}\}$  exactly when  $\mathbf{X}$  has full column rank.

#### Example. (A rank-deficient layout)

Suppose  $n$  observations fall into  $g$  groups and  $\mathbf{X} = [\mathbf{1}, \mathbf{z}_1, \dots, \mathbf{z}_g]$ , where  $\mathbf{z}_k$  is the indicator of group  $k$ . Since  $\mathbf{z}_1 + \dots + \mathbf{z}_g = \mathbf{1}$ , the vector  $(1, -1, \dots, -1)\mathbf{T}$  lies in  $\mathcal{N}(\mathbf{X})$ . The  $g$  indicators are linearly independent (their supports are disjoint), so  $\text{rank}(\mathbf{X}) = g$  while  $p = g + 1$ . By [Eq. 6.2.1](#),  $\mathcal{N}(\mathbf{X})$  is exactly the line spanned by  $(1, -1, \dots, -1)\mathbf{T}$ . The model space  $\mathbb{C}(\mathbf{X}) = \mathbb{C}([\mathbf{z}_1, \dots, \mathbf{z}_g])$  is the set of vectors that are constant within groups.

### 6.2.2 Inner products, orthogonality and complements

Unless stated otherwise  $\mathbb{R}^n$  carries the Euclidean inner product  $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}\mathbf{T}\mathbf{v}$  and norm  $\|\mathbf{u}\| = (\mathbf{u}\mathbf{T}\mathbf{u})^{1/2}$ . Vectors are **orthogonal**, written  $\mathbf{u} \perp \mathbf{v}$ , if  $\mathbf{u}\mathbf{T}\mathbf{v} = 0$ . The Cauchy–Schwarz inequality  $|\mathbf{u}\mathbf{T}\mathbf{v}| \leq \|\mathbf{u}\|\|\mathbf{v}\|$ , with equality iff one vector is a multiple of the other, lets us define the angle  $\theta \in [0, \pi]$  between nonzero vectors by

$$\cos \theta = \frac{\mathbf{u}^T \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}. \quad (6.2.2)$$

The angle comes back in [Section 6.7](#), where  $R^2$  turns out to be a squared cosine. The Pythagorean identity

$$\mathbf{u} \perp \mathbf{v} \implies \|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 \quad (6.2.3)$$

follows by expanding  $(\mathbf{u} + \mathbf{v})^T(\mathbf{u} + \mathbf{v})$ . The cross terms are the ones that vanish.

### Definition 6.2.1 — Orthogonal complement

The **orthogonal complement** of a subspace  $\mathcal{S} \subseteq \mathbb{R}^n$  is

$$\mathcal{S}^\perp = \{\mathbf{v} \in \mathbb{R}^n : \mathbf{v}^T \mathbf{u} = 0 \text{ for all } \mathbf{u} \in \mathcal{S}\}.$$

It is immediate that  $\mathcal{S}^\perp$  is a subspace and that  $\mathcal{S} \cap \mathcal{S}^\perp = \{\mathbf{0}\}$ : a vector orthogonal to itself has zero length. To test whether  $\mathbf{v} \in \mathcal{S}^\perp$  it is enough to check orthogonality to a spanning set of  $\mathcal{S}$ . In particular,  $\mathbf{v} \in \mathbb{C}(\mathbf{A})^\perp$  iff  $\mathbf{A}^T \mathbf{v} = \mathbf{0}$ .

### Lemma 6.2.1

For any  $n \times p$  matrix  $\mathbf{A}$ ,

$$\mathcal{N}(\mathbf{A}^T) = \mathbb{C}(\mathbf{A})^\perp \quad \text{and} \quad \mathcal{N}(\mathbf{A}) = \mathbb{C}(\mathbf{A}^T)^\perp.$$

*Proof.*  $\mathbf{v} \in \mathcal{N}(\mathbf{A}^T)$  means  $\mathbf{A}^T \mathbf{v} = \mathbf{0}$ , that is,  $\mathbf{v}$  is orthogonal to every column of  $\mathbf{A}$ , which is the same as  $\mathbf{v} \perp \mathbb{C}(\mathbf{A})$ . The second identity is the first applied to  $\mathbf{A}^T$ . ■

## 6.2.3 Orthonormal bases and Gram–Schmidt

Vectors  $\mathbf{u}_1, \dots, \mathbf{u}_k$  are **orthonormal** if  $\mathbf{u}_i^T \mathbf{u}_j$  is 1 for  $i = j$  and 0 otherwise. Orthonormal vectors are linearly independent: taking the inner product of  $\sum_j a_j \mathbf{u}_j = \mathbf{0}$  with  $\mathbf{u}_i$  gives  $a_i = 0$ . The coordinates of a vector  $\mathbf{v} \in \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_k)$  are simply the inner products  $a_i = \mathbf{u}_i^T \mathbf{v}$ , with no linear system to solve. This is why orthonormal bases are so convenient.

### Proposition 6.2.2 — Gram–Schmidt

Let  $\mathbf{v}_1, \dots, \mathbf{v}_k$  be linearly independent. Define recursively

$$\mathbf{w}_j = \mathbf{v}_j - \sum_{i < j} (\mathbf{u}_i^T \mathbf{v}_j) \mathbf{u}_i, \quad \mathbf{u}_j = \mathbf{w}_j / \|\mathbf{w}_j\|, \quad j = 1, \dots, k.$$

Then each  $\mathbf{w}_j \neq \mathbf{0}$ . The vectors  $\mathbf{u}_1, \dots, \mathbf{u}_k$  are orthonormal, and  $\text{span}(\mathbf{u}_1, \dots, \mathbf{u}_j) =$

$\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$  for every  $j$ .

*Proof.* Induction on  $j$ . Suppose the claims hold for  $j - 1$ . The vector  $\mathbf{w}_j$  is  $\mathbf{v}_j$  minus an element of  $\text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{j-1}) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{j-1})$ . If  $\mathbf{w}_j$  were zero,  $\mathbf{v}_j$  would lie in that span, contradicting independence. For  $l < j$ ,  $\mathbf{u}_l^T \mathbf{w}_j = \mathbf{u}_l^T \mathbf{v}_j - \sum_{i < j} (\mathbf{u}_i^T \mathbf{v}_j) \mathbf{u}_l^T \mathbf{u}_i = \mathbf{u}_l^T \mathbf{v}_j - \mathbf{u}_l^T \mathbf{v}_j = 0$ , so  $\mathbf{u}_j$  is orthogonal to its predecessors. Finally,  $\mathbf{u}_j \in \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{j-1}, \mathbf{v}_j) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$  and  $\mathbf{v}_j \in \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_j)$ , so the two spans agree. ■

Two consequences are used repeatedly. Every subspace has an orthonormal basis (apply Gram–Schmidt to any basis). And any orthonormal basis of a subspace  $\mathcal{S}$  can be extended to an orthonormal basis of  $\mathbb{R}^n$ : extend it to a basis of  $\mathbb{R}^n$  in any way, then apply Gram–Schmidt, which leaves the first vectors unchanged. [Section 6.10](#) shows that Gram–Schmidt, written in matrix form, is the QR factorization.

## 6.2.4 Direct sums and the orthogonal decomposition

For subspaces  $\mathcal{S}, \mathcal{T}$ , the sum  $\mathcal{S} + \mathcal{T} = \{\mathbf{u} + \mathbf{v} : \mathbf{u} \in \mathcal{S}, \mathbf{v} \in \mathcal{T}\}$  is a subspace. When  $\mathcal{S} \cap \mathcal{T} = \{\mathbf{0}\}$  the sum is *direct*, written  $\mathcal{S} \oplus \mathcal{T}$ . Every element then decomposes as  $\mathbf{u} + \mathbf{v}$  in exactly one way: if  $\mathbf{u} + \mathbf{v} = \mathbf{u}' + \mathbf{v}'$  then  $\mathbf{u} - \mathbf{u}' = \mathbf{v}' - \mathbf{v}$  lies in both subspaces and so is zero. When in addition  $\mathcal{S} \perp \mathcal{T}$ , the sum is an *orthogonal* direct sum.

### Theorem 6.2.3 — Orthogonal decomposition

For every subspace  $\mathcal{S} \subseteq \mathbb{R}^n$ ,

$$\mathbb{R}^n = \mathcal{S} \oplus \mathcal{S}^\perp, \quad \dim \mathcal{S} + \dim \mathcal{S}^\perp = n, \quad (\mathcal{S}^\perp)^\perp = \mathcal{S}.$$

Hence every  $\mathbf{y} \in \mathbb{R}^n$  can be written uniquely as  $\mathbf{y} = \mathbf{u} + \mathbf{v}$  with  $\mathbf{u} \in \mathcal{S}$  and  $\mathbf{v} \perp \mathcal{S}$ .

*Proof.* Let  $\mathbf{u}_1, \dots, \mathbf{u}_r$  be an orthonormal basis of  $\mathcal{S}$ , extended to an orthonormal basis  $\mathbf{u}_1, \dots, \mathbf{u}_n$  of  $\mathbb{R}^n$ . A vector  $\mathbf{y} = \sum_{i=1}^n a_i \mathbf{u}_i$  is orthogonal to  $\mathcal{S}$  iff  $a_1 = \dots = a_r = 0$ , so  $\mathcal{S}^\perp = \text{span}(\mathbf{u}_{r+1}, \dots, \mathbf{u}_n)$  has dimension  $n - r$ . Writing  $\mathbf{y} = \sum_{i \leq r} a_i \mathbf{u}_i + \sum_{i > r} a_i \mathbf{u}_i$  exhibits the decomposition, and it is unique because  $\mathcal{S} \cap \mathcal{S}^\perp = \{\mathbf{0}\}$ . By the same argument with the roles of the two index sets swapped,  $(\mathcal{S}^\perp)^\perp = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_r) = \mathcal{S}$ . ■

Combining [Theorem 6.2.3](#) with [Lemma 6.2.1](#) gives a matrix fact used in almost every later chapter.

### Corollary 6.2.4

For any  $n \times p$  matrix  $\mathbf{A}$ :

- (a)  $\mathcal{N}(\mathbf{ATA}) = \mathcal{N}(\mathbf{A})$ ;
- (b)  $\mathbb{C}(\mathbf{ATA}) = \mathbb{C}(\mathbf{AT})$  and  $\mathbb{C}(\mathbf{AAT}) = \mathbb{C}(\mathbf{A})$ ;
- (c)  $\text{rank}(\mathbf{ATA}) = \text{rank}(\mathbf{AAT}) = \text{rank}(\mathbf{A})$ ;
- (d)  $\mathbb{C}(\mathbf{A})^\perp = \mathcal{N}(\mathbf{AT})$  and  $\mathbb{R}^p = \mathbb{C}(\mathbf{AT}) \oplus \mathcal{N}(\mathbf{A})$ .

*Proof.*

- (a) If  $\mathbf{Ab} = \mathbf{0}$  then  $\mathbf{ATAb} = \mathbf{0}$ . Conversely,  $\mathbf{ATAb} = \mathbf{0}$  gives  $\mathbf{bTATAb} = \|\mathbf{Ab}\|^2 = 0$ , so  $\mathbf{Ab} = \mathbf{0}$ .
- (b) By [Lemma 6.2.1](#) applied to the symmetric matrix  $\mathbf{ATA}$ , and then (a) and [Lemma 6.2.1](#) again,  $\mathbb{C}(\mathbf{ATA}) = \mathcal{N}(\mathbf{ATA})^\perp = \mathcal{N}(\mathbf{A})^\perp = (\mathbb{C}(\mathbf{AT})^\perp)^\perp = \mathbb{C}(\mathbf{AT})$ , using  $(\mathcal{S}^\perp)^\perp = \mathcal{S}$  in the last step. The second identity is the first applied to  $\mathbf{AT}$ .
- (c) Take dimensions in (b) and use  $\text{rank}(\mathbf{AT}) = \text{rank}(\mathbf{A})$ .
- (d) This is [Lemma 6.2.1](#) and [Theorem 6.2.3](#).

■

Part (b) is the reason the normal equations  $\mathbf{XTXb} = \mathbf{XTy}$  can always be solved: the right-hand side lies in  $\mathbb{C}(\mathbf{XT}) = \mathbb{C}(\mathbf{XTX})$ . We come back to this in [Section 6.4](#).

### Lemma 6.2.5

For subspaces  $\mathcal{S}_1, \mathcal{S}_2 \subseteq \mathbb{R}^n$ ,

$$(\mathcal{S}_1 + \mathcal{S}_2)^\perp = \mathcal{S}_1^\perp \cap \mathcal{S}_2^\perp \quad \text{and} \quad (\mathcal{S}_1 \cap \mathcal{S}_2)^\perp = \mathcal{S}_1^\perp + \mathcal{S}_2^\perp.$$

*Proof.* A vector is orthogonal to  $\mathcal{S}_1 + \mathcal{S}_2$  iff it is orthogonal to every  $\mathbf{u}_1 + \mathbf{u}_2$ , which (taking  $\mathbf{u}_2 = \mathbf{0}$  or  $\mathbf{u}_1 = \mathbf{0}$ ) is the case iff it is orthogonal to both subspaces. For the second identity, apply the first to  $\mathcal{S}_1^\perp$  and  $\mathcal{S}_2^\perp$ :  $(\mathcal{S}_1^\perp + \mathcal{S}_2^\perp)^\perp = \mathcal{S}_1 \cap \mathcal{S}_2$ , and take complements of both sides.

■

[Lemma 6.2.5](#) is what we need to describe reduced models defined by linear constraints ([Theorem 6.5.2](#)).

## 6.2.5 Exercises

### 6.2.5.1 A. Check your understanding

#### Exercise A1.

Show that  $\mathcal{S}^\perp$  is a subspace for any set  $\mathcal{S}$  (not necessarily a subspace), and that  $\mathcal{S} \subseteq \mathcal{T}$  implies  $\mathcal{T}^\perp \subseteq \mathcal{S}^\perp$ . Show that  $\mathcal{S}^\perp = \text{span}(\mathcal{S})^\perp$ .

### 6.2.5.2 B. Practice

**Exercise B1.**

A one-way layout has three groups of sizes 2, 2, 1 and model matrix  $\mathbf{X} = [\mathbf{1}, \mathbf{z}_1, \mathbf{z}_2, \mathbf{z}_3]$ . Find bases for  $\mathbb{C}(\mathbf{X})$ ,  $\mathcal{N}(\mathbf{X})$ ,  $\mathbb{C}(\mathbf{X}\mathbf{T})$  and  $\mathbb{C}(\mathbf{X})^\perp$ , and check that the dimensions agree with [Eq. 6.2.1](#) and [Theorem 6.2.3](#).

**Exercise B2.**

Show that  $\mathbb{C}(\mathbf{A}\mathbf{B}) \subseteq \mathbb{C}(\mathbf{A})$ , and deduce  $\text{rank}(\mathbf{A}\mathbf{B}) \leq \min\{\text{rank}(\mathbf{A}), \text{rank}(\mathbf{B})\}$ . Show that  $\mathbb{C}(\mathbf{A}\mathbf{B}) = \mathbb{C}(\mathbf{A})$  whenever  $\mathbf{B}$  has full row rank.

**6.2.5.3 C. Going deeper****Exercise C1.**

For subspaces  $\mathcal{S}, \mathcal{T} \subseteq \mathbb{R}^n$ , prove  $\dim(\mathcal{S} + \mathcal{T}) = \dim \mathcal{S} + \dim \mathcal{T} - \dim(\mathcal{S} \cap \mathcal{T})$ . Use it to show that two  $r$ -dimensional subspaces of  $\mathbb{R}^n$  with  $2r > n$  must share a nonzero vector.